

1) You will be given the schematics of the Delhi Art Gallery (Ground Floor). Design a model to install the fewest possible Closed Circuit Cameras covering all hallways and turns. The input would be the coordinates of the points of turn in two-dimension and your program must print those set of points where CCTV cameras are to be installed.

Input Format

Read the total number of turns (n) in the art gallery

Read the n points of representing turns as ordered pairs (x-coordinate, y-coordinate), in the next n lines

Output Format

Print the points as ordered pairs (x-coordinate, y-coordinate), line-by-line where the CCTV cameras are to be installed, in ascending order of x-coordinate (in case of a tie in x-coordinate, arrange in ascending order of y-coordinate)

Sample Input

7

(0,3)

(2,2)

(1,1)

(2,1)

(3,0)

(0,0)

(3,3)

Sample Output

(0,0)

(0,3)

(3,0)

(3,3)

2) Maximum Flow Using Ford-Fulkerson

You are given a directed graph with N vertices and M edges. Each edge has a specified capacity. Your task is to compute the **maximum flow** from a given **source** node S to a **sink** node T using the **Ford-Fulkerson algorithm**, where the augmenting paths must be found using **Depth-First Search (DFS)**.

Input:

- The first line contains three integers N, M, denoting the number of vertices and edges.
- The next M lines each contain three integers u, v, and c, representing a directed edge from node u to node v with capacity c.
- The last line contains two integers S and T, the source and sink nodes.

Output:

- Print a single integer representing the **maximum flow** from SSS to TTT.

Requirements:

1. Implement the **Ford-Fulkerson method**.
2. Use **DFS** to find augmenting paths in the residual graph.

Sample Input

```
4 5
0 1 10
0 2 5
1 2 15
1 3 10
2 3 10
0 3
```

Output:

```
15
```